DB Browser for SQLite - C:\Users\you\Documents\CharlesRiver_Work.sqlite

New Database | Open Database | Write Changes | Revert Changes

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Open SQL file(s) | Save SQL file | ▶ Execute all/selected SQL

Ex 10.4.sql

```
121 -- Requirement (4): amount invoiced against orders and variance by year, 2024 to 2026
122 -- Population: invoice lines with a purchase-order line, dated 2024 to 2026; expected: 2024 0.48%, 2025 0.47%, 2026 0.51%
123 SELECT SUBSTR(pi.InvoiceDate, 1, 4) AS InvoiceYear,
124     COUNT(*) AS InvoiceLines,
125     ROUND(SUM(pil.Quantity * pil.UnitCost), 2) AS AmountInvoiced,
126     ROUND(SUM(pil.LineTotal), 2) AS LineTotals,
127     ROUND(SUM(pil.Quantity * (pil.UnitCost - pol.UnitCost)), 2)
128         AS PriceVariance,
129     ROUND(100.0 * SUM(pil.Quantity * (pil.UnitCost - pol.UnitCost))
130         / SUM(pil.Quantity * pil.UnitCost), 2) AS VariancePct
131 FROM PurchaseInvoiceLine AS pil
132     INNER JOIN PurchaseInvoice AS pi
133         ON pi.PurchaseInvoiceID = pil.PurchaseInvoiceID
134     INNER JOIN PurchaseOrderLine AS pol
135         ON pol.POLineID = pil.POLineID
136 WHERE pi.InvoiceDate BETWEEN '2024-01-01' AND '2026-12-31'
137 GROUP BY InvoiceYear
138 ORDER BY InvoiceYear;
```

	InvoiceYear	InvoiceLines	AmountInvoiced	LineTotals	PriceVariance	VariancePct
1	2024	6898	13491998.13	13491998.93	64631.92	0.48
2	2025	13529	17468202.42	17468203.84	82377.43	0.47
3	2026	19866	21612260.34	21612262.33	109566.2	0.51

Execution finished without errors.
Result: 3 rows returned
At line 123:
SELECT SUBSTR(pi.InvoiceDate, 1, 4) AS InvoiceYear,

Outlined: VariancePct, 0.48%, 0.47% and 0.51%, a near-steady share of what is invoiced against orders while that amount grows 60% from 2024 to 2026. The variance grows with the amount invoiced. AmountInvoiced mixes quantity, order prices and mix, so the comparison cannot say how much of the growth is more buying. LineTotals differs from the amount by the rounding of each line.